

1 What really happens in a laser

2 Bridging the gap between a real laser and a water tank

3 Simulating the water

Variables and their meanings:

1. H : average water level (where the tank is filled to)
2. h : wave height (the real water height is $h+H$)
3. u : horizontal water velocity in the x-direction
4. v : horizontal water velocity in the y-direction
5. P_{ext} : external pressure applied to the water surface

Shallow Water Equations:

$$\frac{\partial h}{\partial t} + \frac{\partial(H+h)u}{\partial x} + \frac{\partial(H+h)v}{\partial y} = 0 \quad (1)$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + g \frac{\partial h}{\partial x} + ku + \frac{1}{\rho} \frac{\partial P_{ext}}{\partial x} = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (2)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + g \frac{\partial h}{\partial y} + kv + \frac{1}{\rho} \frac{\partial P_{ext}}{\partial y} = \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \quad (3)$$

Note: P_{ext} is the only term which drives the water. Additionally, ν (the kinematic viscosity) is very small, ρ (water's density) $\approx 1000 \frac{kg}{m^3}$, and k (the viscous drag coefficient) ≈ 1 for water.

3.1 Updating the fields through time

To simulate the differential equations, the differentials can be approximated to tiny intervals.

For the terms of the form $\frac{\partial f}{\partial t}$ for some f , they were used to update the field f from one frame to the next. i.e. using $\{f(t), f(t - \Delta t), f(t - 2\Delta t), \dots\}$ to determine $f(t + \Delta t)$.

$$\frac{\partial f}{\partial t} \approx \frac{\Delta f}{\Delta t} = \frac{f(t + \Delta t) - f(t)}{\Delta t} \quad (4)$$

But only using one piece of (potentially noisy) information, $f(t)$, makes the simulation really susceptible to exploding.

To make it more stable, $f(t - \Delta t)$ will also be used.

$$\frac{\partial f}{\partial t} \approx \frac{3s_1 + 1s_2}{4} \quad (5)$$

With s_1 the slope of f from the current frame to the last, and s_2 the slope from the last frame to the one before it. Therefore

$$\frac{\partial f}{\partial t} \approx \frac{3 \cdot \frac{f(t+\Delta t) - f(t)}{\Delta t} + 1 \cdot \frac{f(t) - f(t-\Delta t)}{\Delta t}}{4} = \frac{h(t + \Delta t) - \frac{2}{3}h(t) - \frac{1}{3}h(t - \Delta t)}{\frac{4}{3}\Delta t} \quad (6)$$

Given an equation of the form

$$\frac{\partial f}{\partial t} = A \quad (7)$$

$$\frac{h(t + \Delta t) - \frac{2}{3}h(t) - \frac{1}{3}h(t - \Delta t)}{\frac{4}{3}\Delta t} \approx A \quad (8)$$

The recursive formula to update f is

$$h(t + \Delta t) = \frac{4}{3}\Delta t \cdot A + \frac{2}{3}h(t) + \frac{1}{3}h(t - \Delta t) \quad (9)$$

3.2 Approximating spacial derivatives

The spacial derivatives can be approximated in a similar way, this time taking two slopes on each side.

$$\frac{\partial f}{\partial x} = \frac{s_1 + 3s_2 + 3s_3 + s_4}{8} \quad (10)$$

$$\frac{\partial f}{\partial x} = \frac{-f(x - 2\Delta x) - 2f(x - \Delta x) + 0f(x) + 2f(x + \Delta x) + f(x + 2\Delta x)}{8\Delta x} \quad (11)$$

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3.3 Implementing boundaries in the simulation

The simulation features reflective boundaries. To do this, the horizontal velocity near the borders was forced to gradually become 0 with distance to the border. Additionally, if values would have been sampled outside of the simulation's borders, the point gets reflected across the border and back into the simulated area by however much it would have "overshot".

4 Electrons in the field

Since the water surface is mostly flat with vertical waves, in modeling the electric field we'll consider it as two-dimensional (in the xy plane), and assume its vector at each point in space is pointing in the up-down direction.

We imagine an electron as a particle around an atom. We imagine the attraction to the nucleus as a restoring force in the form $-kz$ (like a spring force). We also imagine that the atom is in the plane of the field ($z = 0$). Being a charged particle with charge $-e$, it is affected by the electric field $\mathbf{E}(\mathbf{r}, t)$ at its position $\mathbf{r} = (x, y)$ and at time t . Finally, we add a small damping force $F_D = -\beta\dot{\mathbf{r}}$, accounting for energy lost anywhere other than light emission.

Since the electric field is modeled to always be in the up-down ($\hat{z}$) direction, the electron's oscillation will only happen in that direction.

Writing Newton's second law for the electron:

$$F_{net} = m\ddot{z} \quad (12)$$

$$-kz - \beta\dot{z} + qE(\mathbf{r}, t) = m\ddot{z} \quad (13)$$

$$\frac{q}{m}E = \ddot{z} + \frac{\beta}{m}\dot{z} + \frac{k}{m}z \quad (14)$$

This is the form of a driven and damped harmonic oscillator. It is not so simple, though, because the electron, being a charged particle, affects the electric field as it is moved. For example, a wave pulse comes by in the up ($+z$) direction. This leads the electron to get pushed down (since its charge is negative). The electron is now below the 2d plane of the electric field². Then, since it's a negatively charged particle, it pulls the field toward it, downward.

4.1 Absorption

Therefore, the electron can be seen to effectively damp the electric field around it. This then damps its own motion.

While there does exist a real effect similar to this called "radiative damping", this is not how it really works. We ended up with a similar result because we did something strange, assuming that the field is effectively two-dimensional with all vectors pointing in the z -direction. Then, paradoxically, we imagine the electron to possibly be above or below the field, but that it is still affected by the field, from the point just above or below its position.

The damping effect which was just described, although it is not the real way that it works, can be thought of as the absorption of light by a ground-state (normal, not excited) electron. Absorption is conveniently solved for free using this setup.

For example, in the case that the electric field is in the form $E = \sin(x)$, the electron will be driven by the term $\frac{q}{m}E$. Since $q < 0$, it will begin oscillating with a motion like $z = -\sin(x)$.³ Then the electron pulls on the field,

¹ $f(x)$ will always have a coefficient of 0 whenever the approximation is taken with equal weighting on the left and right of the point. This was deemed acceptable, as the approximation being symmetric was assumed to be more important.

² but we still imagine that the electron is affected by the nearest point on the field, even if it is technically not inside the field anymore

³ The electron will not necessarily oscillate at the same frequency as the field, this depends on the choice of the "spring constant" k . This will be discussed later

so near the electron, the field becomes something like $E + z = \sin(x) - \sin(x) = 0$. The light near the electron disappears, it is absorbed.

4.2 Emission

The other important feature of an electron in a laser is emission: producing light. Complicated quantum mechanics would be necessary for a physically accurate description. For simplicity, this will be ignored. Instead, emission will be mimicked by using the following observation: if an electron pulling the field towards it damps the field looks like it absorbs the light, then pushing the field away from it will look like the electron is emitting light.

More precisely, it will make the electron add to the amplitude of whatever wave is passing through it. It is actually important that it does this, as opposed to simply generating a new wave which is unrelated to what is happening near the electron. This is because it ensures constructive interference. Without the assurance of constructive interference, the electron might add nothing to the field, or otherwise destructively interfere with it.

Mathematically, this effect amounts to "flipping the sign" of the electron's effect on the field. We can think of it as the field seeing an electron as having a positive charge and reacting accordingly, while it still moves as if it has a negative one.⁴

4.3 When to absorb, when to emit

Absorption and emission of light from an atom depends on the atom's "excitement". When excited (this is practically done by applying a current to the atoms), the electrons jump to higher energies, and so when a wave passes by them (when they get "stimulated") they release that energy from the excitement into the wave. If they aren't already excited and they're in "ground state", the electrons might use the energy in that wave to jump to higher energy themselves. Note that this is assuming a two-level model of the electrons (the levels being "excited" and "ground" as described).

For now, we will assume that all atoms are clumped together such that the electric field looks exactly the same near all atoms. This simplifies calculations.

The technique of adding energy to an atom's electrons is called "pumping". We'll label the pumping rate (the number of electrons put into the excited state per second) as R . Electrons which are excited also tend to naturally lose energy (for example, in the form of heat) and drop to the ground state. We'll label the time it takes for an electron to naturally drop to the ground state, the "relaxation time" as T . Finally, a wave passing through a ground state electron excites it at a rate proportional to the wave's intensity I . Similarly, a wave passing through an excited electron tends to make it drop to the ground state at a rate proportional to I .

Letting N_E be the number of excited electrons and N_G be the number of ground-state electrons in the system, we can write the above ideas into a differential equation giving the rate at which the number of excited electrons increases.

$$\dot{N}_E = R - \frac{N_E}{T} - \alpha I N_E + \alpha I N_G \quad (15)$$

Note that we have the nearby wave intensity I as if it's the same for all electrons.

Let $N = N_E - N_G$, which we can call the net excitement. Also, let $N_T = N_E + N_G$, the total number of electrons in the system, which doesn't change. These variables allow us to simply express the situation in terms of one changing quantity, the net excitement N .

We can rewrite Equation 15 as

$$\dot{N} + \left(\frac{1}{T} + 2\alpha I\right)N = 2R - \frac{N_T}{T} \quad (16)$$

To get a sense of how this equation acts, we can assume that the light intensity will be constant and solve for N as a function of time t . The initial net excitement will be called N_0 .

$$N(t) = \left(N_0 - \frac{2R - \frac{N_T}{T}}{\frac{1}{T} + 2\alpha I}\right) e^{-(\frac{1}{T} + 2\alpha I)t} + \frac{2R - \frac{N_T}{T}}{\frac{1}{T} + 2\alpha I} \quad (17)$$

Note that as $t \rightarrow \infty$, $N \rightarrow \frac{2R - \frac{N_T}{T}}{\frac{1}{T} + 2\alpha I}$. This indicates that the system has a behavior where it comes to rest at an equilibrium.

⁴Note that if the electron began to move as if it had a positive charge too, it would just be a positively charged particle while still absorbing light.

You may have noticed that N is a count and it therefore only has integer values, while Equation 17 has $N(t)$ as a continuous function. To solve this inconsistency, let's take Equation 16 and divide both sides by N_T , then redefine R so that it now represents $\frac{R}{N_T}$. This new R now represents the pumping rate per atom. Letting $n = \frac{N}{N_T}$, Equation 16 now becomes

$$\dot{n} + \left(\frac{1}{T} + 2\alpha I\right)n = 2R - \frac{1}{T} \quad (18)$$

If N_T is large enough, n could take any value between -1 and 1 and therefore it can be thought of as approximately continuous.

A useful term which is commonly used is "population inversion", when there are more excited atoms than ground-state atoms. In other words, when $n > 0$.

4.3.1 Calculating wave intensity I at any point and time

Equation 18 requires I being known, the intensity of the wave at a specific position and time. We must calculate this for a water wave, since the end goal is to reproduce the effect in water.

To calculate I at a certain instant and a certain position, we use the identity $I = \frac{dE}{dV} ||\mathbf{u}||$, where $\frac{dE}{dV}$ is the energy density and $||\mathbf{u}||$ is the speed of the wave at the desired position.

Let's calculate dE of the wave, the energy at a particular element in the water tank.

$$dE = dE_K + dE_p \quad (19)$$

$$dE_K = \frac{1}{2} dm \mathbf{u}^2 \quad (20)$$

$$dE_P = dm g z \quad (21)$$

Since the density of water is known, we can say $dm = \rho dV$

$$dE = \frac{1}{2} \rho dV \mathbf{u}^2 + \rho dV g z \quad (22)$$

$$I = \frac{\frac{1}{2} \rho dV \mathbf{u}^2 + \rho dV g z}{dV} ||\mathbf{u}|| \quad (23)$$

We're using a simplifying approximation and thinking about each vertical column of water a whole element, with the same velocity $\mathbf{u}$. We'll continue the tradition by integrating the above equation across the vertical direction, from 0 at the bottom of the tank to $h + H$, the total water height. Now, instead of each point in 3D space having its own intensity, simply each vertical column of water has its own intensity. Expanding dV with $dV = dx dy dz$:

$$I = \frac{\frac{1}{2} \rho dx dy dz \mathbf{u}^2 + \rho dx dy dz g z}{dx dy dz} ||\mathbf{u}|| \quad (24)$$

$$I = \frac{\frac{1}{2} \rho dz \mathbf{u}^2 + \rho dz g z}{dz} ||\mathbf{u}|| \quad (25)$$

$$I \approx \frac{\int_{z=0}^{z=h+H} \left[\frac{1}{2} \rho dz \mathbf{u}^2 + \rho dz g z \right]}{\int_{z=0}^{z=h+H} dz} ||\mathbf{u}|| \quad (26)$$

$$I \approx \frac{\frac{1}{2} \rho (h + H) \mathbf{u}^2 + \frac{1}{2} \rho (h + H)^2 g}{h + H} ||\mathbf{u}|| \quad (27)$$

$$I \approx \frac{1}{2} \rho \mathbf{u}^3 + \frac{1}{2} \rho (h + H) g ||\mathbf{u}|| \quad (28)$$

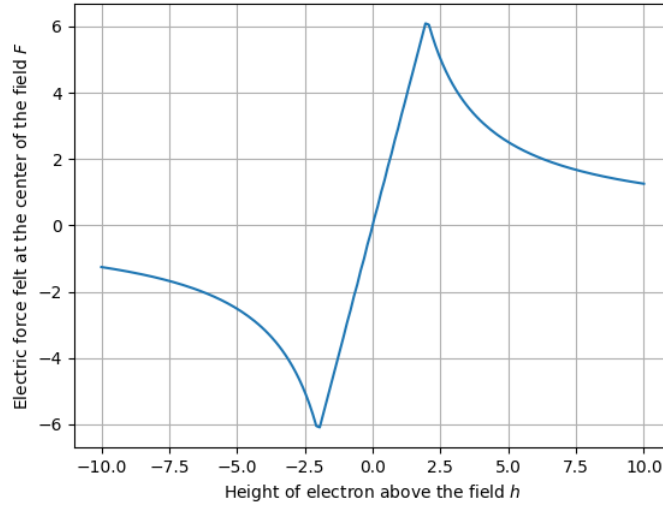


Figure 1: Graph

4.4 An electron's force on the field

We would like to quantify how an electron pushes on the field.

When discussing an electric field and how it interacts with a particle like an electron, the electron's own contribution to the field is usually ignored when analyzing that electron's motion. This is because the field only encodes the forces which would be applied if a particle was there. At the position of a point charge like an electron, there is a singularity, and that's a large problem conceptually. However, we can get around this by treating the electron as a ball with volume. Let's see how the field reacts at the point directly at the center of the electron, as the electron moves up and down around it. Let's do it in 2d for simplicity, with "the field" being a line at h (height) = 0, keeping with how we're treating the field as an object that the electron can go above and below.

For each point inside the electron's volume, it produces a tiny force

$$d\mathbf{F} = -\frac{k dq}{r^2} \hat{\mathbf{r}} \quad (29)$$

We've been ignoring the horizontal components of the force, so let's just look at the vertical part

$$dF = \frac{k dq}{x^2 + z^2} z \quad (30)$$

Where x and z are the horizontal and vertical distances to this point within the electron. We must now add up all of the points to get the effect of the electron as a whole. C denotes that we're integrating over the circle encapsulating the electron.

$$F = \int \int_C \frac{k dq}{x^2 + z^2} z \quad (31)$$

Let's say that the electron has radius R and charge q , then $\frac{q}{R} = \frac{dq}{dA}$, so in polar coordinates, $dq = \frac{q}{R} r dr d\theta$.

$$F(h) = \int_0^{2\pi} \int_0^R \frac{q}{R} \frac{k r}{x^2 + z^2} z dr d\theta \quad (32)$$

Written in terms of r and θ , $x = r \cos \theta$ and $z = r \sin \theta + h$, where h is the height of the center of the electron relative to the field.

Integrating numerically and writing the force as a function of the height of the electron, we get Figure 1. It can be seen that while the electron is still overlapping the origin ($h < R$), the force felt is linear.

This function looks similar to a function of the form

$$F(h) = \frac{A}{h^2 + R^2} h \quad (33)$$

Where R is the radius of the electron

4.5 How the electrons use n

In the proper quantum mechanics way of describing atoms in a laser, when population inversion is attained $n > 0$, the atoms have a net constructive effect on the laser's light in a similar way to how we described emission. Then when $n < 0$, the atoms have a net destructive effect in the same way we described absorption.

As it was described in "Absorption" and "Emission", an absorbing "ground state" electron acts as one would expect from an electron in an electric field. An emitting "excited" electron has an effect on the field with a flipped sign, ie pushing the field away from the electron's position. Where is this sign exactly? It's in Equation 29. An absorbing electron would have the negative sign, an emitting one wouldn't.

So if all electrons were in ground state ($n = -1$) the cluster of N_T electrons would, as a whole follow Equation 29. If they were all in the excited state ($n = 1$), the cluster would produce a force like Equation 29 but without the negative sign. Then if half were absorbing and half were emitting, their individual contributions would cancel out to produce the force $d\mathbf{F} = 0$. In general, we have the cluster producing a force:

$$d\mathbf{F} = n \frac{k dq}{\mathbf{r}^2} \hat{\mathbf{r}} \quad (34)$$

4.6 The choice of k and its impact

At the beginning of this section, we said that the electrons are attracted to the nucleus by a restoring force $F = -kx$. This value is important as it determines the colour of the laser.

Lasers usually only emit a single wavelength of light. This is because the electrons in the lasing medium will only respond well to oscillations with a frequency near their natural frequency, $\omega_o := \sqrt{\frac{k}{m}}$.

More precisely,